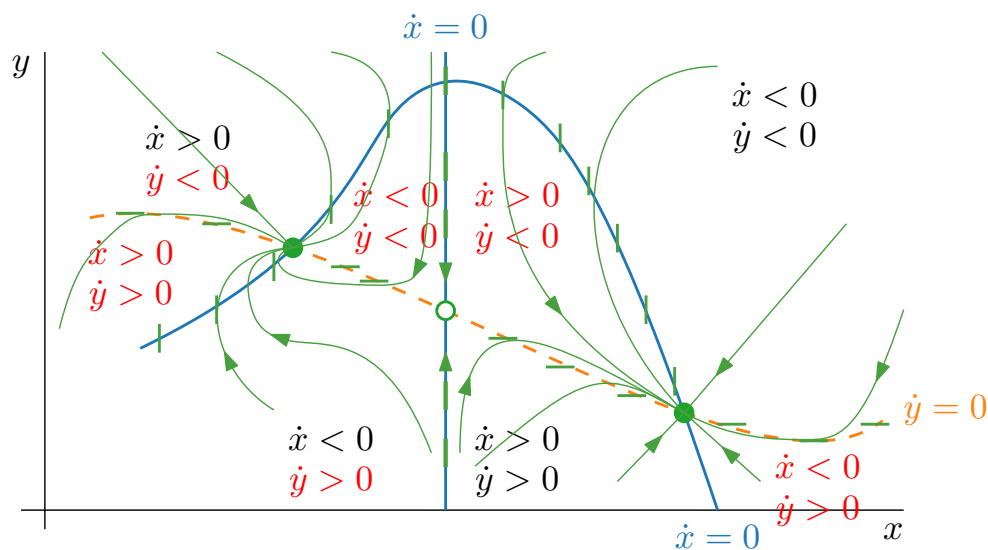


PROBLEMS CLASS 2: SOLUTIONS

1. Phase portrait recipe

Trajectories in a phase portrait do not intersect, because where they intersect, the trajectory (i.e. value of $(\dot{x}, \dot{y})$) should be unique in a deterministic (i.e. non-random) system.

2. Trajectory drawing practice



3. Bifurcation investigation (Exam 2023)

For each value of k , there are three values of x_0 :

$$x_0 = 0, \quad k, \quad \text{and } 1.$$

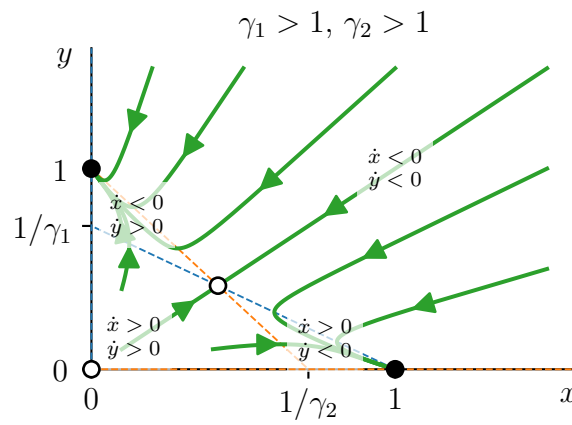
This suggests an equation of the form

$$\frac{dx}{dt} = -x(x - k)(x - 1),$$

i.e. the Allee model seen in class. Drawing the portrait confirms the stability.

4. Competitive Lotka–Volterra

- Equilibria only exist where $\dot{x} = 0$ and $\dot{y} = 0$ lines intersect.
- Completed diagram looks like:



5. Problem sheet 1: variations

The nullclines for $\dot{x}$ and $\dot{y}$ are

$$y = \frac{x+10}{2} \quad \text{and} \quad y = -x - 2, \quad (1)$$

and the phase plane looks like

